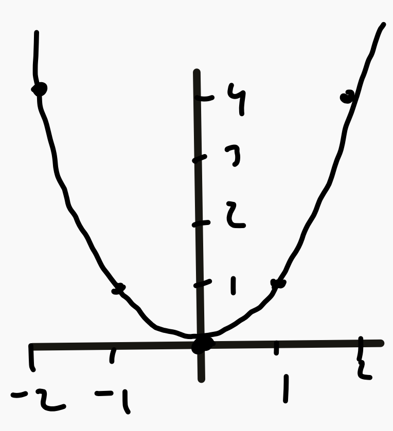
$y = x^2$

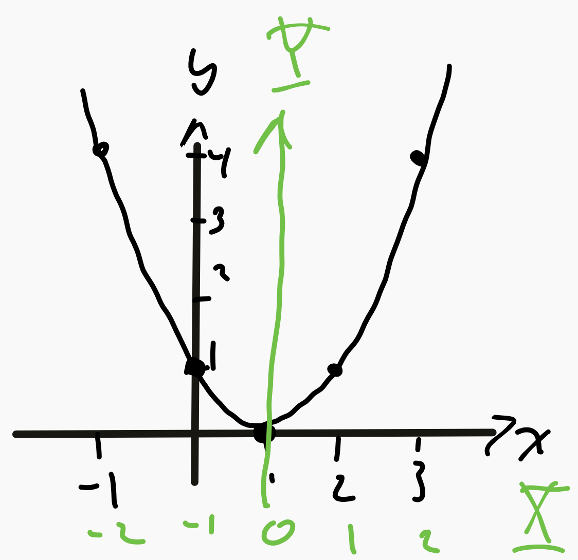


x	x^2
-2	4
-1	1
0	0
1	1
2	4

right shift by 1

$y = (x-1)^2$

x	$x-1$	$(x-1)^2$
-2	-3	9
-1	-2	4
0	-1	1
1	0	0
2	1	1
3	2	4



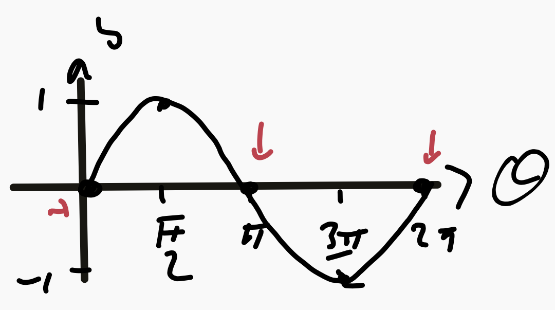
$X = x-1$

$Y = X^2$

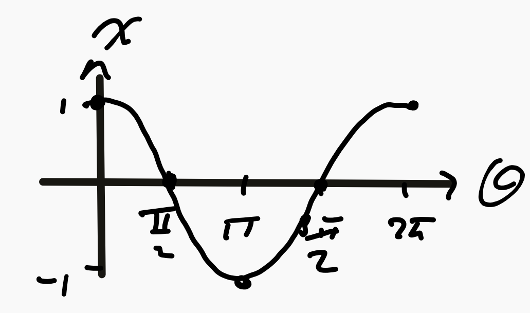
$x-1=0$

$x=1$

$y = \sin\theta$



$x = \cos\theta$



2) $y = \sin(\overset{1}{t} - \frac{\pi}{4})$

amp = 1   flip: no

period = $\frac{9\pi}{4} - \frac{\pi}{4} = \frac{8\pi}{4} = 2\pi$

phase shift = $\frac{\pi}{4}$

y=0

$t - \frac{\pi}{4} = 0$, $t - \frac{\pi}{4} = \pi$, $t - \frac{\pi}{4} = 2\pi$

$t = \boxed{\frac{\pi}{4}}$ $t = \frac{4\pi}{4} + \frac{\pi}{4} = \frac{5\pi}{4}$ $t = \frac{8\pi}{4} + \frac{\pi}{4} = \frac{9\pi}{4}$

y=1

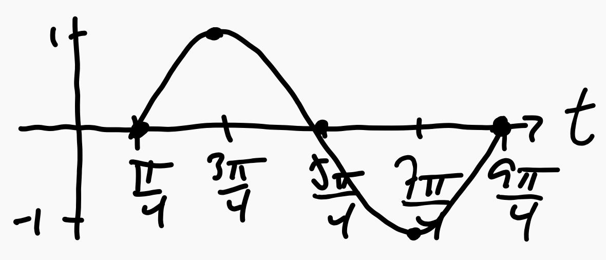
$t - \frac{\pi}{4} = \frac{\pi}{2}$

$t = \frac{2\pi}{4} + \frac{\pi}{4} = \frac{3\pi}{4}$

y=-1

$t - \frac{\pi}{4} = \frac{3\pi}{2}$

$t = \frac{6\pi}{4} + \frac{\pi}{4} = \frac{7\pi}{4}$



recip

2 period: $2\pi(\frac{1}{2}) = \pi$

3) $y = 5\sin(2t - \frac{\pi}{3})$

amp: 5   flip: no

period: $\frac{7\pi}{6} - \frac{\pi}{6} = \frac{6\pi}{6} = \pi$

phase shift = $\frac{\pi}{6}$

y=0

$2t - \frac{\pi}{3} = 0$, $2t - \frac{\pi}{3} = \pi$, $2t - \frac{\pi}{3} = 2\pi$

$\frac{1}{2}(2t) = (\frac{\pi}{3})\frac{1}{2}$ $2t = \frac{3\pi}{3} + \frac{\pi}{3}$ $2t = \frac{6\pi}{3} + \frac{\pi}{3}$

$t = \boxed{\frac{\pi}{6}}$ $\frac{1}{2}(2t) = \frac{\overset{2}{\cancel{4}}\pi}{3}(\frac{1}{\cancel{2}})$ $\frac{1}{2}(2t) = \frac{7\pi}{3}(\frac{1}{2})$

$t = \frac{2\pi}{3}$ $t = \frac{7\pi}{6}$

y=1

$2t - \frac{\pi}{3} = \frac{\pi}{2}$

$2t = \frac{\pi}{2} + \frac{\pi}{3}$

$= \frac{3\pi + 2\pi}{6}$

$\frac{1}{2}(2t) = \frac{5\pi}{6}(\frac{1}{2})$

$t = \frac{5\pi}{12}$

y=-1

$6(2t - \frac{\pi}{3}) = (\frac{3\pi}{2})6$

$12t - 2\pi = 9\pi$

$12t = 11\pi$

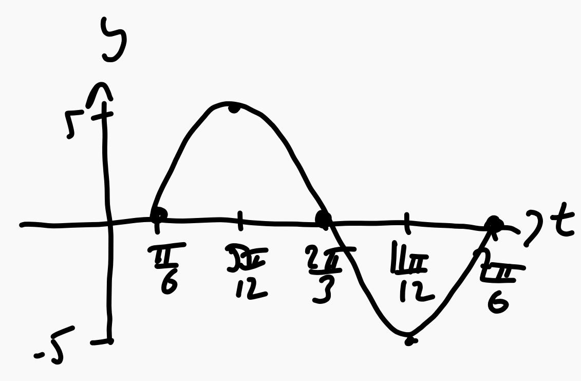
$t = \frac{11\pi}{12}$

$6(2t - \frac{\pi}{3}) = (\frac{\pi}{2})6$

$12t - 2\pi = 3\pi$

$12t = 5\pi$

$t = \frac{5\pi}{12}$



reciprocal

$\frac{1}{2}$ period $2\pi(\frac{2}{1}) = 4\pi$

4) $x = 2\cos(\frac{1}{2}t - \frac{5\pi}{6})$

amp: 2   flip: no

period: $\frac{17\pi}{3} - \frac{5\pi}{3} = \frac{12\pi}{3} = 4\pi$

phase shift: $\frac{5\pi}{3}$

x=1

$6(\frac{1}{2}t - \frac{5\pi}{6}) = 0(6)$, $6(\frac{1}{2}t - \frac{5\pi}{6}) = 2\pi(6)$

$3t - 5\pi = 0$ $3t - 5\pi = 12\pi$

$3t = 5\pi$ $3t = 17\pi$

$t = \boxed{\frac{5\pi}{3}}$ $t = \frac{17\pi}{3}$

x=0

$6(\frac{1}{2}t - \frac{5\pi}{6}) = \frac{\pi}{2}(6)$, $6(\frac{1}{2}t - \frac{5\pi}{6}) = \frac{3\pi}{2}(6)$

$3t - 5\pi = 3\pi$ $3t - 5\pi = 9\pi$

$3t = 8\pi$ $3t = 14\pi$

$t = \frac{8\pi}{3}$ $t = \frac{14\pi}{3}$

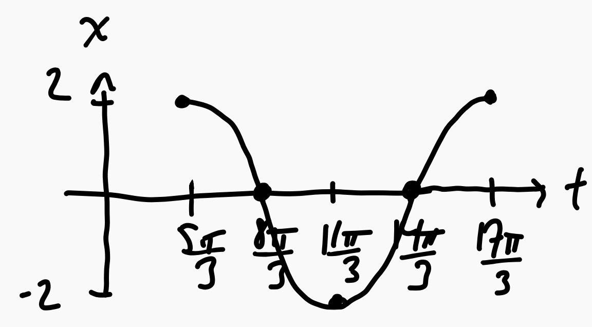
x=-1

$6(\frac{1}{2}t - \frac{5\pi}{6}) = \pi \cdot 6$

$3t - 5\pi = 6\pi$

$3t = 11\pi$

$t = \frac{11\pi}{3}$



recipal

1 period $2\pi(\frac{1}{1})$

5) $y = \sin(t + \frac{\pi}{2})$

amp = 1   flip: no

period = $\frac{3\pi}{2} - (-\frac{\pi}{2}) = \frac{3\pi}{2} + \frac{\pi}{2} = 2\pi$

phase shift = $-\frac{\pi}{2}$

y=0

$t + \frac{\pi}{2} = 0$, $t + \frac{\pi}{2} = \pi$, $t + \frac{\pi}{2} = 2\pi$

$t = \boxed{-\frac{\pi}{2}}$ $t = \frac{2\pi}{2} - \frac{\pi}{2} = \frac{\pi}{2}$ $t = \frac{4\pi}{2} - \frac{\pi}{2} = \frac{3\pi}{2}$

y=1

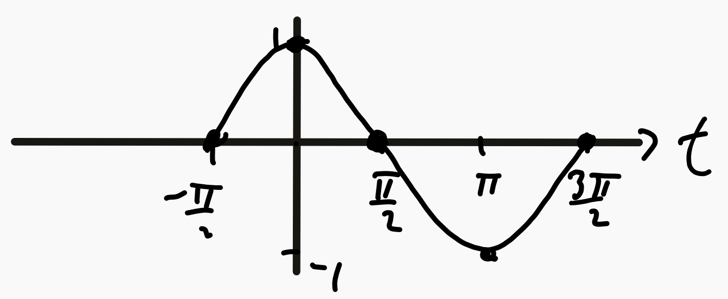
$t + \frac{\pi}{2} = \frac{\pi}{2}$

$t = \frac{\pi}{2} - \frac{\pi}{2} = 0$

y=-1

$t + \frac{\pi}{2} = \frac{3\pi}{2}$

$t = \frac{3\pi}{2} - \frac{\pi}{2} = \pi$



recip

π period: $2\pi(\frac{1}{\pi}) = 2$

6) $x = \cos(\pi t + \frac{\pi}{4})$

amp = 1   flip: no

period = $\frac{7}{4} - (-\frac{1}{4}) = \frac{8}{4} = 2$

phase shift: $-\frac{1}{4}$

x=1

$\pi t + \frac{\pi}{4} = 0$, $\cancel{\pi}t + \frac{\cancel{\pi}}{4} = 2\cancel{\pi}$

$\pi t = -\frac{\pi}{4}$ $t = 2 - \frac{1}{4}$

$t = \boxed{-\frac{1}{4}}$ $= \frac{8}{4} - \frac{1}{4} = \frac{7}{4}$

x=0

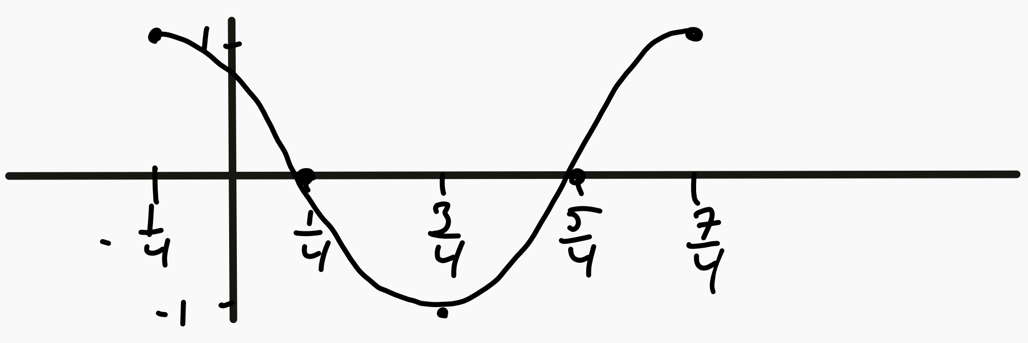
$\cancel{\pi}t + \frac{\cancel{\pi}}{4} = \frac{\cancel{\pi}}{2}$, $\cancel{\pi}t + \frac{\cancel{\pi}}{4} = \frac{3\cancel{\pi}}{2}$

$t = \frac{2}{4} - \frac{1}{4} = \frac{1}{4}$ $t = \frac{6}{4} - \frac{1}{4} = \frac{5}{4}$

x=-1

$\cancel{\pi}t + \frac{\cancel{\pi}}{4} = \cancel{\pi}$

$t = \frac{4}{4} - \frac{1}{4} = \frac{3}{4}$



7) period, phase shift

$y = 5\sin(2t - \frac{\pi}{3})$

period: $\frac{7\pi}{6} - \frac{\pi}{6} = \frac{6\pi}{6} = \pi$

phase shift: $\frac{\pi}{6}$

$2t - \frac{\pi}{3} = 0$

$2t = \frac{\pi}{3}$

$t = \frac{\pi}{6}$

$2t - \frac{\pi}{3} = 2\pi$

$2t = \frac{6\pi}{3} + \frac{\pi}{3} = \frac{7\pi}{3}$

$t = \frac{7\pi}{6}$

$y = \genfrac{}{}{0pt}{}{\sin}{\cos}(B\theta + C)$

$B\theta + C = 0$ $B\theta + C = 2\pi$

$B\theta = -C$ $B\theta = 2\pi - C$

$\theta = \frac{-C}{B}$ $\theta = \frac{2\pi}{B} - \frac{C}{B}$

period: $\frac{2\pi}{B} \cancel{- \frac{C}{B} - (-\frac{C}{B})} = \frac{2\pi}{B}$

recip

$2\pi(\ )$

phase shift: $-\frac{C}{B}$